Assignment 1 – Figures, illustrations and tables

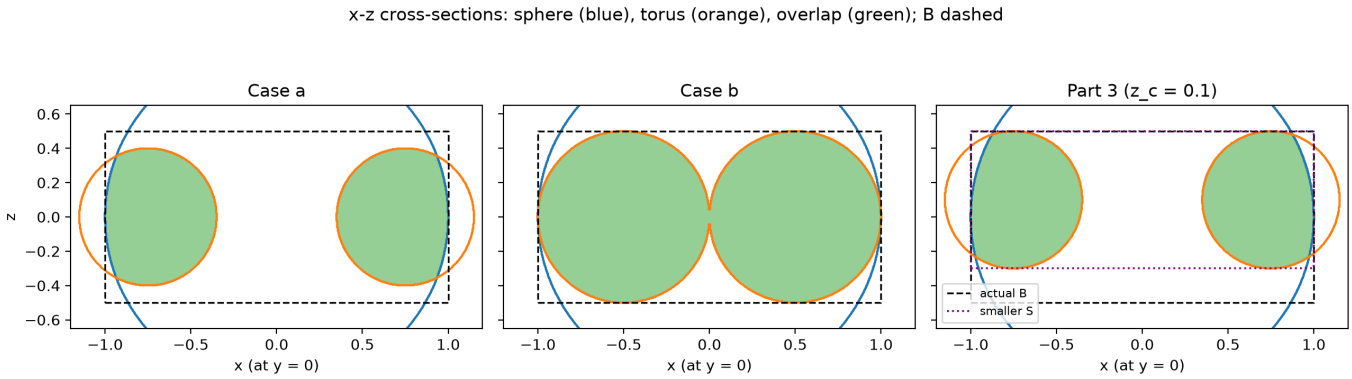


Figure 1: x–z cross-sections at $y = 0$. Green: intersection. Dashed: box $B = [-1, 1]^2 \times [-0.5, 0.5]$; dotted: $S = [-1, 1]^2 \times [-0.3, 0.5]$.

- how much space the sphere and the torus share (the green region).
- our method: scatter random points evenly through the dashed box, count the fraction that land in the green region, and multiply by the box volume (4).
- The box holds the green region in every case, including the shifted torus.

Table 1: Uniform Monte Carlo, $N = 10^5$, box B ($V_B = 4$). Reference: independent 1D integration.

Part	Parameters	$\hat{V}$	95% CI ±	Reference	Error
1a	$k = 1, R = 0.75, r = 0.4$	1.8572	0.0124	1.8561	+0.0011
1b	$k = 1, R = 0.5, r = 0.5$	2.4701	0.0120	2.4674	+0.0027
3a	as 1a, centre $(0, 0, 0.1)$	1.8210	0.0123	1.8284	-0.0074

- with 100,000 points, case a gives 1.857 and case b gives 2.470, each give or take 0.012 (95% CI)
- validation: we also worked out the volume a second way, by slicing the shapes into thin horizontal layers and adding them up: 1.856 and 2.467.
- case b has an exact answer, $\frac{\pi^2}{4}$, because the whole torus fits inside the sphere.
- both check values fall inside our ranges.

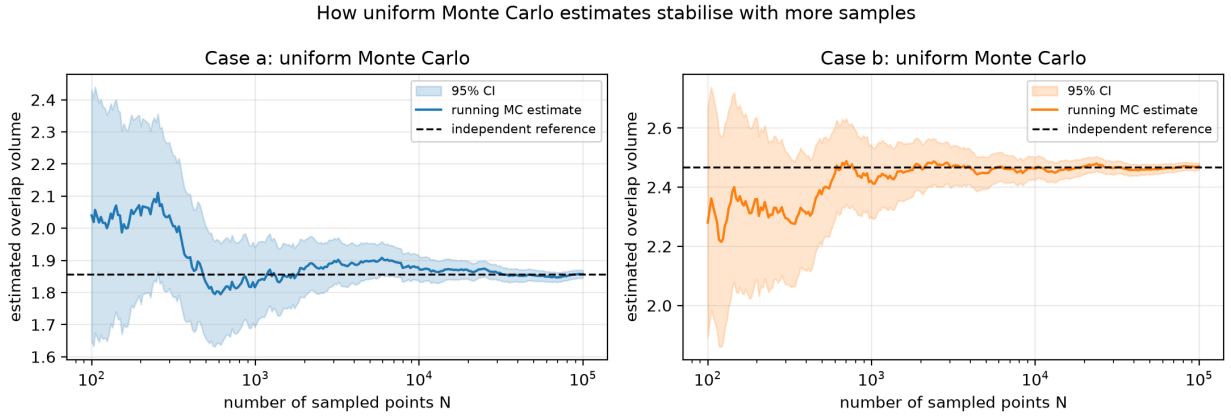


Figure 2: Running uniform MC estimate with 95% CI band (Part 1 samples).

- shows the estimate as points are added: it jumps around early, then settles, and the range narrows.
- x4 the sample size approximately halves the mc standard error. (the actual estimation error need not decrease after every additional sample)

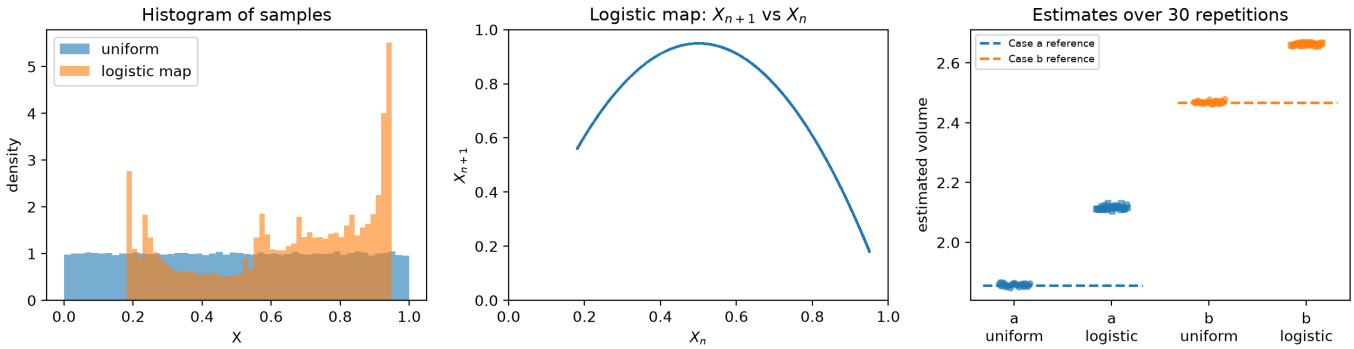


Figure 3: Part 2: logistic map ($m = 3.8$) vs. uniform. Left: histogram. Middle: lag-1 plot. Right: estimates over 30 runs; dashed: reference.

- replaces the random numbers with the logistic map
- left shows it is not a fair replacement. the numbers never go below (approx.) 0.18 or above 0.95, and they pile up near 0.95. So some parts of the box get no points and others get too many.
- middle plots each number against the next. they all sit on one curve, so each number is fixed by the one before. The result is 2.12 instead of 1.86, and 2.66 instead of 2.47.
- thirty different starting values give almost the same wrong answer, so it looks precise but is consistently off.
- "The raw logistic map generates nonuniform, dependent samples. Increasing the sample size may reduce fluctuations, but does not generally remove the resulting bias in the uncorrected volume estimator. The standard i.i.d. Monte Carlo error formula is therefore not justified."

Table 2: Part 2: 30 runs of $N = 10^5$ (uniform: 30 seeds; logistic: 30 X_0 triples).

Case	Sampler	Mean $\hat{V}$	SD over runs	Mean error	Error/$(s/\sqrt{30})$
a	uniform	1.8578	0.0049	+0.0017	1.9
a	logistic	2.1165	0.0060	+0.2605	237.8
b	uniform	2.4693	0.0044	+0.0019	2.4
b	logistic	2.6627	0.0048	+0.1953	224.1

- moves the torus up by 0.1. We get 1.821, give or take 0.012, against a check value of 1.828.
- shared volume drops by about 1.5 percent, because the torus moves away from the sphere's widest part and a little more of it sticks out.
- Independent numerical integration shows that shifting the torus upward by 0.1 reduces the intersection volume by approximately 1.49%. The Monte Carlo estimates show a 1.95% decrease, consistent with sampling uncertainty

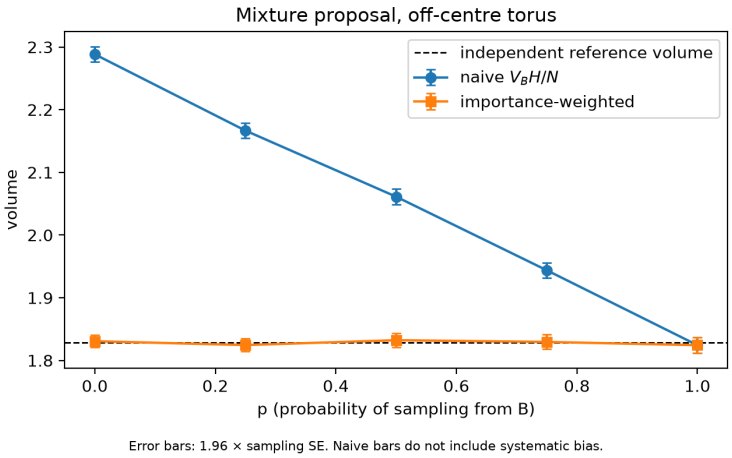


Figure 4: Part 3b: mixture $p\mathcal{U}(B) + (1-p)\mathcal{U}(S)$. Error bars: 95% sampling CI; dashed: reference 1.8284.

- takes a point from the full box with probability p, and otherwise from a smaller box S around the torus.
- with the same counting rule, the answer rises as p falls, up to 2.29 when every point comes from S.
- flaw: the rule assumes every part of the box gets an equal share of points, but now points crowd into S, so the shape collects too many hits.
- Importance weighting corrects the bias caused by the nonuniform mixture proposal. The weighted estimates fluctuate around the reference volume, and all five are consistent with it within their approximate 95% confidence intervals.
- Sampling from S alone reduces the theoretical variance by approximately 36.8% compared with uniform sampling from B. This is possible because S contains the entire intersection. Using the original box-volume multiplier without correction, however, produces a biased estimate.

Table 3: Part 3b, $N = 10^5$ per p (independent runs). $E[\hat{V}_{\text{naive}}] = V_{\text{ref}}(p + 1.25(1-p))$.

p	$\hat{V}_{\text{naive}}$	$E[\hat{V}_{\text{naive}}]$	$\hat{V}_w$	95% CI $\pm$
0.00	2.2883	2.2855	1.8307	0.0098
0.25	2.1666	2.1712	1.8245	0.0104
0.50	2.0612	2.0569	1.8322	0.0110
0.75	1.9438	1.9427	1.8295	0.0117
1.00	1.8244	1.8284	1.8244	0.0123

- evenly spread random points work
- their error ranges can be trusted
- numbers that only look random give a confident but wrong answer
- if you sample some areas more often, you must correct for it.